

School of Arts and Sciences
Department of Computer Science and Information Technology
Summer 2024– 2025
CSCI380 Software Engineering

Class Day	:	_____	Class Time:	_____	Room:	_____
Instructor	:	_____				
E-mail	:	_____				
Office	:	_____	Office Hours :	_____	Extension No:	_____

I. Course Description

Introduction to the fundamental concepts of software engineering, focusing on the processes, methodologies, tools, and techniques used in the development of large-scale software systems.

Topics include problem analysis, requirements gathering and analysis, system architectural design and modeling processes, coding standards, project management, and maintaining quality throughout the software development lifecycle. Through a combination of lectures, practical exercises, and hands-on projects, students will explore deeply various software development models such as the waterfall model, and design and modeling models, processes and techniques. Students will learn how to analyze a problem and develop the appropriate analysis and design models to formalize the requirements using object-oriented methods and appropriate theory.

II. Course Objectives

In a **Software Engineering course**, the objectives are designed to ensure that students gain both theoretical knowledge and practical skills in the development and maintenance of software systems.

CO-1: Familiarize students with the stages of the SDLC, including requirements gathering, design, implementation, testing, deployment, and maintenance.

CO-2: Teach students how to collect, analyze, and define software requirements in collaboration with stakeholders.

CO-3: Develop the ability to analyze and break down real-world problems into smaller, manageable tasks. Ensure students understand software design principles, including modularization, abstraction, and separation of concerns.

CO-4: Introduce students to project management methodologies and help them develop skills for working in teams.

CO-5: Enable students to apply their knowledge in practical projects and real-world scenarios through case studies and capstone projects.

III. Course Learning Outcomes

- CLO-1:** Gain knowledge and ability to select and apply appropriate development models (Waterfall, RAP, etc.) for different projects.
- CLO-2:** Ability to document functional and non-functional requirements, as well as create use cases and user stories.
- CLO-3:** Ability to design scalable, maintainable, and efficient software architectures using design patterns like client/server and architectural frameworks.
- CLO-4:** Ability to plan, track progress, and collaborate effectively within a development team, meeting deadlines and milestones.
- CLO-5:** Ability to analyze and break down real-world problems into smaller, manageable tasks. Ability to design, develop, and deliver a complete software product, integrating all aspects of the SDLC.

IV. Mapping to program outcomes (PLOs are listed at the end of the syllabus)

Computer Science Overview	PLO-1	PLO -2	PLO -3	PLO -4	PLO -5	PLO -6	PLO -7	PLO -8	PLO -9
CLO-1		X							
CLO-2		X					X		
CLO-3	X	X			X				
CLO-4						X			
CLO-5				X					X

V. Topics Covered

The course covers a wide array of topics designed to give students a comprehensive main understanding of how software is developed. The course blends theory with practical application, often culminating in a hands-on capstone project to demonstrate the students' ability to apply their knowledge in real-world scenarios.

- Introduction to Software Engineering – Overview, Role of Software Engineers
- Software Development Life Cycle (SDLC) – Models and phases
- Requirements Engineering – Gathering, Specification, Use Cases and User Stories
- Software Design – Principles, Architectural Styles, Data modeling, User Interface (UI) modeling, UML - Use case diagrams, class diagrams, sequence diagrams, etc.
- Software Project Management – Planning and Tracking
- Capstone Project - Real-World Application, Presentation and Documentation

VI. Prerequisite – Corequisite

CSCI335 Database Systems

ENGL201 Composition and Research Skills

VII. Course Material: Book or handouts

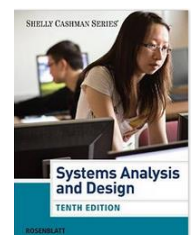
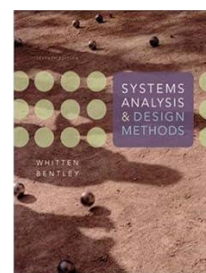
Systems Analysis and Design Methods (7th Edition) by Witten, Bentley and Dittman. Publisher: McGraw-Hill/Irwin. ISBN: 73052337

Oreilly – Learning UML (soft copy is available)

Further Reading(s)

Systems Analysis and Design by Harry J. Rosenblatt. Publisher: Cengage Learning; 10 edition (March 1, 2013) ISBN-10

Software(s) required - MS Project – Visio - Power Designer or similar software



VIII. Course Assessment

Activity	Percentage
Project	15%
Test	20%
Midterm	30%
Final Exam	35%

IX. Detailed Weekly Schedule

Week		Book Reference	Topic Details
Week 1	Session 1 30/6	Book reference & handout material	Syllabus: reading and explaining -Overview of the course - Overview of software engineering - Historical overview/Software crisis! - Introduction to the software development lifecycle (SDLC)
	Session 2 1/7	Book reference	Chapter 1: Systems Analysis and Stakeholders - Define Systems Analysis and Design main terminology. - Define stakeholders and the system development process.
	Session 3 2/7	Handout material	Software Development Life Cycle (SDLC) Models – overview (focus on some models, comparison study): - Re-iterate/Rapid overview; system development process Waterfall Model, Rapid Application Development (RAD) Model / Prototype model Incremental Model - Iterative Model, Spiral Model
	Session 4 3/7	Handout material	Software Development Life Cycle (SDLC) Models – Incremental Model - Iterative Model, Spiral Model - For reading-Extra materials; V-Model.
Week2	Session 1 7/7	Handout material & Book reference (Lect1.2 to explore)	Chapter 2: Software Development Life Cycle (SDLC) - Requirement Analysis - What is Systems Analysis, Systems Analysis Approaches? - Define system requirements, differentiate between functional and nonfunctional requirements, the activity of problem analysis, Requirements management stakeholders - Lect1.2>> rapid explore of relevant topics
	Session 2 8/7	Book reference	Chapter 2: Software Development Life Cycle (SDLC) - Requirement Analysis Introduction to Joint requirements planning (JRP) session Discussing problem covering chapter 2
	Session 3 9/7	Handout material	Chapter 3: Project Management What's Project Management, Project Management Life Cycle? >> Gantt Chart, Pert Chart
	Session 4 10/7	Handout material	Chapter 3: Project Management >> Exercices on Gantt / Pert Charts

Week 3	Session 1 14/7		Chapter 4: Feasibility Analysis Feasibility Analysis and the System Proposal, Tests for Feasibility, Cost-Benefit Analysis Techniques >> Exercise on Feasibility Analysis
	Session 2 15/7		>> Extra-exercises on Gantt / Pert Charts
	Session 3 16/7	Midterm Exam Review	Revisions and Exercises
	Session 4 17/7	Midterm exam Thursday 17 July	Chapters 1, 2,3 and 4
Week 4	Session 1 21/7		<u>Chapter 5: Modeling System requirement using Use Case</u> - Introduction to Use Case Modeling. >> Use case modeling diagram. Solving problem covering chapter 5/ Use Case Modeling
	Session 2 22/7		<u>Chapter 5: Modeling System requirement with Use Cases</u> Solving problem covering chapter 5/ Use case narratives
	Session 3 23/7		<u>Chapter 6: System/Software Architectural design</u> Introduction to Software Architectural design, Software components - Software Architectural styles - Client/Server, etc.
	Session 4 24/7		<u>Chapter 6: System/Software Architectural design</u> Software components – Backend (data containers)/Frontend (GUI forms) >> design/modeling techniques and CAISE- tools Project: Part II of the project is due Use case Analysis
Week 5	Session 1 28/7	Test and Review	Test during class time duration 45 mins <u>Chapter 6: System/Software Architectural design</u> • Design-Patterns, an overview – relevance!
	Session 2 29/7		<u>Chapter 7: Process Modeling</u> • Introduction to Systems Modeling, • Mapping System/Software Requirements to System/Software Architecture. Process Modeling, the Process of Logical Process Modeling Solving problem covering chapter 7
	Session 3 30/7		Project: Part III of the project is due: • 3-System Architecture – Detailed design: Client/Server style: • Back-End ; Data modeling – ERDiagram/ClassDiagram >> CASE tools like PowerDesigner (Data modeling/ClassDiagram, ...) • Front-End – GUI Forms, ...>> CASE tools like Figma (Interface modeling). Project: >> Final Project Presentations

	Session 4 31/7		Project: >> Final Project Presentations)
Final Exam: All chapters			

Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam’s grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

E-mails

Students should use their university e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your university e-mail accounts.

University Peer Review Policy

All classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4		72	2.2	
65	1.5	D+	73	2.3	
66	1.6		74	2.4	
75	2.5	C+	83	3.3	B+
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9		87	3.7	
80	3	B	88	3.8	A
81	3.1		89	3.9	
82	3.2		>=90	4	

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	

Computer Science Outcomes

Program outcomes	Program Outcomes
category -1 Underlying Conceptual Basis for Informatics	PLO -1 Describe and explain the essential facts, concepts, theories and mathematical methods relevant to computing, computing equipment, computer communication and informatics applications as appropriate to their program of study
	<i>PLO-1 for CSIT=> Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software and their practical application</i>
category -2 Analysis	PLO -2 Select and use relevant analytic, modelling and simulation methods
	PLO-3 Choose appropriate solution patterns, algorithms and data structures
category -3 Design and Implementation	PLO -4 Describe the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts
	<i>PLO-4 for CSIT=>Specify and design computing/network hardware/software which meet specified requirements</i>
category -4 Economic, legal, social, ethical and environmental context	PLO -5 Explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring and use of informatics-based systems
category -5 Informatics practice	PLO -6 Describe and explain management techniques appropriate to the design, implementation, testing, deployment and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques
	<i>PLO -6for CSIT=>Demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>
category -6 Other Professional Skills and Competences	PLO -7 Communicate effectively both verbally and using a variety of communications media
	PLO -8 Plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development
	PLO -9 Participate effectively in informatics group-working